

# Lucy Martinez

Department of Mathematics, Rutgers University

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🖱 Website: marti310.github.io

## EDUCATION

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|-----------------------------------------------------|---------------|
| <b>Ph.D. Mathematics</b> , Rutgers University       | Expected 2027 |
| Advisor: Doron Zeilberger                           |               |
| <b>M.S. Mathematics</b> , Rutgers University        | May 2023      |
| Awarded upon completion of written qualifying exams |               |
| <b>B.S. Mathematics</b> , Stockton University       | May 2021      |
| Minor in Computer Science                           |               |

## RESEARCH INTERESTS

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Combinatorics, parking functions, permutation properties, probability, game theory, random walk

## FELLOWSHIPS, GRANTS, AND AWARDS

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### Fellowships

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|--------------------------------------------------------------------------------|-------------|
| California State University Chancellor's Doctoral Incentive Program Fellowship | 2026 - 2027 |
| Rutgers School of Arts and Sciences Excellence Fellowship                      | 2026 - 2027 |
| Rutgers School of Arts and Sciences Excellence Fellowship                      | 2022 - 2023 |
| Rutgers School of Graduate Studies Dean's Fellowship                           | 2021 - 2022 |
| National Science Foundation Graduate Research Fellowship (NSF-GRFP)            | 2021 - 2026 |

### Research and Travel Grants

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|-----------------------------------------------------------------------------|---------------|
| Joel Lebowitz Research Fellowship, Rutgers University (\$2,500)             | July 2026     |
| Society for Industrial and Applied Mathematics Student Travel Award (\$650) | 2026          |
| SGS Research & Travel Award, Rutgers University (\$1,000)                   | 2025 - 2026   |
| Collaborate@ICERM Program Award (\$850)                                     | January 2026  |
| MAA Tensor Women & Mathematics Grants: EDGE Webinar (\$5,947.20)            | 2025 - 2026   |
| Joel Lebowitz Research Fellowship, Rutgers University (\$2,500)             | July 2025     |
| Travel Award, Association for Women in Mathematics (\$2,000)                | May 2025      |
| Society for Industrial and Applied Mathematics Student Travel Award (\$650) | 2025          |
| SGS Research & Travel Award, Rutgers University (\$1,000)                   | 2024 - 2025   |
| Society for Industrial and Applied Mathematics Student Travel Award (\$800) | 2024          |
| Dr. Sylvia T. Bozeman Predoctoral Fellowship (\$350)                        | March 2024    |
| SGS Research & Travel Award, Rutgers University (\$1,000)                   | 2023 - 2024   |
| Foundations of Computational Mathematics Conference Travel Award (\$1,900)  | June 2023     |
| Dr. Sylvia T. Bozeman Predoctoral Fellowship (\$500)                        | June 2023     |
| Stockton University REU Award (\$1,000)                                     | December 2019 |

**Awards, Honors, and Scholarships**

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|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|
| AAECC Best Paper Award of 2022, AAECC (€4,000)                                                                                          | 2023          |
| Awarded for publication #2: <i>On Kostant's weight <math>q</math>-multiplicity formula for <math>\mathfrak{sl}_4(\mathbb{C})</math></i> |               |
| Princeton Prospective PhD Preview Program - Honorific Mention                                                                           | 2020          |
| High Impact Practices Scholarship, Stockton University (\$500)                                                                          | December 2020 |
| Stockton University Foundation Scholarship (\$5,180)                                                                                    | 2019 - 2021   |
| Math Alliance Scholar                                                                                                                   | 2019 - 2021   |
| Hispanic Fund Scholarship                                                                                                               | 2019 - 2021   |

**PUBLICATIONS**

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**Preprint**

15. *A parking function analog of the Schröder numbers*. L. Martinez and D. Zeilberger. arXiv: (2026).
14. *Automatic Enumeration of Tilings by Polyominoes*. L. Martinez. arXiv:2607.29369 (2026).
13. *Lehmer parking functions and their outcomes*. M. Beerbower, J. Elder, P. E. Harris, I. Lavene, L. Martinez, A. Martinson, and M. Oldham. arXiv:2603.20535 (2026).
12. *How many coin tosses would you need until you get  $n$  Heads or  $m$  Tails?*. S. Janson, L. Martinez, and D. Zeilberger. arXiv:2512.07803 (2025).
11. *Enumerating vector parking functions and their outcomes based on specified lucky cars*. M. Ferreri, P. E. Harris, L. Martinez and E. Swartz. arXiv:2508.13917 (2025).

**Accepted/In Press**

10. *Lucky cars in Fubini rankings and unit Fubini rankings*. C. Barreto, M. Beerbower, J. Elder, P. E. Harris, L. Martinez, J. L. Ramírez, S. Ramírez, G. Shirley, and J. C. Vásquez. To appear in *Studia Scientiarum Mathematicarum Hungarica*. arXiv:2510.27574.
9. *Parking functions with a fixed set of lucky cars*. P. E. Harris and L. Martinez. To appear in the *Bulletin of the Institute of Combinatorics and its Applications*. arXiv:2410.08057.

**Published**

8. *A symbolic computational approach to the generalized gambler's ruin problem in one and two dimensions*. L. Martinez. *Journal of Difference Equations and Applications*, Vol. 32 (2026), no. 3, 387-411.
7. *Primeness of generalized parking functions*. S. Armon, J. Beckford, D. Hanson, N. Krawzik, O. Mandelshtam, L. Martinez, and C. Yan. *The Electronic Journal of Combinatorics*, Vol. 32 (2025), no. 4, P4.48.
6. *Hitting  $k$  primes by dice rolls*. N. Alon, Y. Malinovsky, L. Martinez, and D. Zeilberger. *The Electronic Journal of Combinatorics*, Vol. 32 (2025), no. 4, P4.16.
5. *Pullback parking functions*. J. Elder, P. E. Harris, L. Koene, I. Lavene, L. Martinez, and M. Oldham. *American Journal of Combinatorics*, Vol. 4 (2025), 1-22.

4. *A guide to the risk-averse gambler and resolving the St. Petersburg paradox once and for all.* L. Martinez and D. Zeilberger. The College of Mathematics Journal, Vol. 55 (2024), no. 3, 226-234.
3. *Repetitions of Pak-Stanley labels in  $G$ -Shi arrangements.* C. Bennett, L. Martinez, A. Mock, G. Rojas Kirby, and R. Truax. Discrete Mathematics, Vol. 347 (2024), no. 9.
2. *On Kostant's weight  $q$ -multiplicity formula for  $\mathfrak{sl}_4(\mathbb{C})$ .* R. E. Garcia, P. E. Harris, M. Loving, L. Martinez, D. Melendez, J. Rennie, G. Rojas Kirby, and D. Tinoco. Applicable Algebra in Engineering, Communication and Computing, Vol. 33 (2022), no. 4, 353-418.

## Expository

1. *How many dice rolls would it take to reach your favorite kind of number?* L. Martinez and D. Zeilberger. Maple Transactions, Vol. 3 (2023), no. 3, Article 15953.

## TEACHING EXPERIENCE

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### Institute for Pure & Applied Mathematics

Teaching Assistant, Practicum for Undergraduate MAThematicians April 2024  
 Three 1-hour problem sessions on introductory topics in matroid theory.

### Rutgers University

Instructor, Young Scholars Program in Discrete Mathematics  
 Combinatorics and Probability I Summer 2026  
 Teaching Assistant  
 Calculus II for the Mathematical and Physical Sciences Fall 2025  
 Calculus I for the Mathematical and Physical Sciences Fall 2024  
 Grader, Computer Science Linear Programming Graduate Course Fall 2023  
 Academic Success Mentor Fall 2022 - Spring 2024

### Stockton University

Teaching Assistant  
 Quantitative Reasoning Spring 2018 - Spring 2021  
 Educational Opportunity Fund Summer 2018  
 Assisted students during assigned math modules for a six-week program designated for low-income students.  
 Mathematics Tutor (F20-S21 Head Tutor) Spring 2018 - Spring 2021

## MENTORING EXPERIENCE

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University of Puerto Rico, Ponce Summer 2024  
 Graduate research mentor for the Combinatorics and Coding Theory in the Tropics REU

Brown University, ICERM institute Summer 2022  
 Graduate research mentor for the Summer@ICERM: Computational Combinatorics REU

## INVITED TALKS

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How Many Coin Tosses Would You Need until You Get  $n$  Heads Or  $m$  Tails?, Graduate Student Combinatorics Conference, University of Illinois Chicago, March 27, 2026

Parking Functions with a Fixed Set of Lucky Cars, EDGE Special Session, AWM Research Symposium, May 16, 2025

Primeness of Vector Parking Functions, MRC Algebraic Combinatorics Special Session, Joint Mathematics Meetings, January 9, 2025

The  $G$ -Shi Arrangement: Pak-Stanley Labels and Games on Paths, Special Session on Enumerative Combinatorics, AMS Fall Central Sectional, University of Texas, September 14, 2024

How Many Dice Rolls Would It Take to Hit Your Favorite Kind of Number?, Special Session on The EDGE Program, Joint Mathematics Meetings, January 4, 2024

The Power of Computation, Mathematics and Statistics Colloquium, The College of New Jersey, October 18, 2023

The  $G$ -Shi Arrangement: Pak-Stanley Labels and Games on Paths, Combinatorics Seminar, Berkeley University, March 20, 2023

Minimum Rank of Regular Bipartite Graphs, Student Mathematics Conference, Moravian College, February 13, 2021

Minimum Rank of Regular Bipartite Graphs, Special Session on Research in Mathematics by Undergraduates and Students in Post-Baccalaureate Programs, Joint Mathematics Meetings, January 7, 2021

Minimum Rank of Regular Bipartite Graphs, Algebraic Combinatorics Session, Canadian Mathematical Society, December 5, 2020

Involution Derangement Graphs, Young Mathematicians Conference, The Ohio State University, August 2020

Involution Derangement Graphs, REU Conference, Clemson University, July 2020

Closed Formulas for Kostant's Partition Function for  $\mathfrak{sl}_4(\mathbb{C})$ , Student Mathematics Conference, Moravian College, February 22, 2020

Geometric Constructions for Lie Algebra Type  $A_3$ , Combinatorial Algebra meets Algebraic Combinatorics Conference, Dalhousie University, January 26, 2020

The  $q$ -analog of Kostant's Weight Multiplicity Formula for  $\mathfrak{sl}_4(\mathbb{C})$ , The Simons Laufer Mathematical Sciences Institute (f.k.a MSRI), July 25, 2019

## CONTRIBUTED PRESENTATIONS IN LECTURE FORMAT

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How Many Coin Tosses Would You Need until You Get  $n$  Heads Or  $m$  Tails?, Extremal Combinatorics Contributed Presentation, SIAM Conference on Discrete Mathematics, June 23, 2026

(In Spanish) Parking Functions with a Fixed Set of Lucky Cars, ICMAM Central America and the Caribbean Conference, October 14, 2025

A Symbolic Computational Approach to the Generalized Gambler's Ruin Problem in One Dimension, Numerical Analysis Contributed Presentation, SIAM Annual Meeting, July 29, 2025

From Linear Algebra to Graph Theory: Minimum Rank of Regular Bipartite Graphs, Graphs Contributed Presentation, SIAM Linear Algebra Conference, May 17, 2024

## LOCAL COLLOQUIUM/SEMINAR TALKS

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How Many Coin Tosses Would You Need until You Get  $n$  Heads Or  $m$  Tails?, Experimental Mathematics Seminar, Rutgers University, February 26, 2026

Parking Functions with a Fixed Set of Lucky Cars, Mathematics Seminar, Stockton University, September 15, 2025

Parking Functions with a Fixed Set of Lucky Cars, Graduate Combinatorics Seminar, Rutgers University, April 9, 2025

A Mirror Step Variant of Gambler's Ruin, Experimental Mathematics Seminar, Rutgers University, February 13, 2025

Simulating the Gambler's Ruin Problem, Graduate Research Café, Rutgers University, January 22, 2025

How can we define prime vector parking functions?, Graduate Combinatorics Seminar, Rutgers University, November 20, 2024

The Chip-firing Game and More, Graduate Pizza Seminar, Rutgers University, November 10, 2023

The  $G$ -Shi Arrangement: Pak-Stanley Labels and Games on Paths, Mathematics Seminar, Stockton University, September 25, 2023

The Chip-firing Game and More, Mathematics Seminar, Stockton University, October 17, 2022

Chromatic Symmetric Function of Trees, Mathematics Seminar, Stockton University, November 30, 2020

Geometric Constructions for Lie Algebra Type  $A_3$ , Mathematics Seminar, Stockton University, March 29, 2020

## NON-RESEARCH TALKS

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Mirror, Mirror: A random walk with a twist, Math Circle, Rutgers University, November 20, 2025

Navigating Academia Despite the Obstacles, Our Stories Session, Fourth Annual OURFA<sup>2</sup>M<sup>2</sup> Conference, November 18, 2023

Mentoring Experiences Through the Lens of a Mentee, Critical Issues in Mathematics Education, The Simons Laufer Mathematical Sciences Institute (f.k.a MSRI), March 23, 2023

## POSTERS

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Parking Functions with a Fixed Set of Lucky Cars, AWM Research Symposium, May 16, 2025

Weyl Alternation Sets for  $\mathfrak{sl}_4(\mathbb{C})$ , Joint Mathematics Meetings, January 17, 2020

Weyl Alternation Sets for  $\mathfrak{sl}_4(\mathbb{C})$ , SACNAS Conference, November 1, 2019

## RESEARCH AND WORKSHOP PARTICIPATION

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### Research Visits

|                                                                                                                       |              |
|-----------------------------------------------------------------------------------------------------------------------|--------------|
| Collaborate@ICERM, Brown University                                                                                   | January 2026 |
| Women in Noncommutative Algebra and Representation Theory Workshop 4,<br>Banff International Research Station, Canada | March 2025   |
| Mathematical Research Communities, American Mathematical Society                                                      | June 2024    |

### Summer Schools

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|-----------------------------------------------------------------------------|-----------|
| Applied Math skills Improvement for Grad studies (AMIGAs), IPAM             | July 2023 |
| Algebraic Methods in Combinatorics, University of Illinois Urbana-Champaign | June 2023 |
| Graph Coloring, University of Illinois Urbana-Champaign                     | May 2023  |

### Workshops

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|-----------------------------------------------------------------------|--------------|
| Integrability and Algebraic Combinatorics, IPAM                       | April 2024   |
| Modern Math Workshop, IMSI                                            | October 2023 |
| Foundations of Computational Mathematics, Sorbonne Université, France | June 2023    |
| Modern Math Workshop, MSRI                                            | October 2022 |
| Open Problems in Algebraic Combinatorics, UMN Twin Cities             | May 2022     |
| The Women of Color Project, Harvard University                        | October 2020 |
| IMA/Math Alliance Workshop on Career Paths, UMN Twin Cities           | July 2020    |
| Modern Math Workshop - MSRI                                           | October 2019 |

### Programs

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|---------------------------------------------------------------------------------------------|-----------|
| Mathematics Outreach Seminar and Training,<br>National Museum of Mathematics, New York City | July 2025 |
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Latinas Achieve in STEM (LaSTEM), Stockton University

February - April 2021

This 7-week program was offered to 5 undergraduate Latinas interested in pursuing careers in STEM. One of the projects included a website that is showcased in the Student Transition Programs.

## WRITING CONTRIBUTIONS TO THE PROFESSION

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10. Beyond Math: The Value of Research Collaborations for Career Advancement. K. P. Hadaway, P. E. Harris, K. J. Harry, L. Martinez, M. G. Norris, and R. Patrias. Math Values. May 28, 2026.
9. Empowering future leaders in Applied Mathematics: The AMIGAs Program. T. Azevedo, D. Beckelhymer, J. Jones, D. Galan Covarrubias, L. Martinez, and A. Singhal. SIAM News Blog, November 3, 2023.
8. AMIGAs' Impact: Computation, collaboration, community ( $C^3$ ). T. Azevedo, D. Beckelhymer, J. Jones, D. Galan Covarrubias, L. Martinez, and A. Singhal. AWM Newsletter, Vol. 53, no. 6, November/December 2023, 25-27.
7. Questions to ask to identify a supportive graduate program as a woman or gender minority in math. B. Gelb, K. Beck, J. Hamlin, H. Mahmoudian, N. Malawo, L. Martinez, and R. Whitman. Math Values. April 12, 2023.
6. A model for a one-hour workshop on mentoring. P. E. Harris, Z. Markman, L. Martinez, A. Mock, E. J. Pabón Cancel, A. Verga, and S. Wang. MAA FOCUS, Vol. 41, no. 1, February/March 2023, 18-21.
5. Ten tips to being productive online for students - and everyone. L. Martinez. Math Values. February 25, 2021.
4. Supporting math majors and grad students in the time of pandemic. P. E. Harris, G. Marquez, and L. Martinez. e-Mentoring Network in the Mathematical Sciences-connecting students and mentors. AMS blogs. January 26, 2021.
3. Networking basics for math undergrads. J. Beckford, A. Christensen, P. E. Harris, L. Martinez, E. Torres Davila, and F. O. Ulysse. e-Mentoring Network in the Mathematical Sciences-connecting students and mentors. AMS blogs. July 21, 2020.
2. Learning during the pandemic: What we wish our professors and mentors knew. M. Cortez, M. Gutierrez, P. E. Harris, B. Keene-Gomez, L. Martinez, A. Minino, A. Prieto Langerica, K. Tesfaye, and S. e-Mentoring Network in the Mathematical Sciences-connecting students and mentors. AMS blogs. May 04, 2020.
1. Bank of REU/Grad fair questions. L. Martinez and E. Torres Davila. e-Mentoring Network in the Mathematical Sciences-connecting students and mentors. AMS blogs. January 24, 2019.

## PROFESSIONAL DEVELOPMENT

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### Workshop Leader

How to Create your Professional Website

AWM Chapter, Rutgers University

The University of Puerto Rico at Ponce

March 2025

July 2024

CV Writing Workshop  
AWM Chapter, Rutgers University  
The University of Puerto Rico at Ponce

November 2024/December 2025  
July 2024

## Panelist

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| Graduate Fellowship Student Panel, Rutgers University                  | September 2026 |
| A Day in the Life of a Mathematician, National Museum of Mathematics   | March 2026     |
| Undergraduate and Graduate Students, IGEN-Math                         | February 2025  |
| Online Fall Graduate School Fair Kickoff Panel, AMS                    | October 2024   |
| Current and Recent Graduate Student Perspectives, GROW Conference      | September 2024 |
| Stockton University Alumni Panel                                       | September 2024 |
| Funding and Fellowships Panel, The Women+ of Color Project             | October 2022   |
| Graduate School Panel, Summer@ICERM                                    | June 2022      |
| Undergraduate Research Student Panel, CURM                             | June 2022      |
| Beginning Undergraduates' Mathematical research Preparation Panel, YSU | May 2022       |
| Student Panel, IMA/Math Alliance Workshop, University of Minnesota     | July 2020      |
| Educational Opportunity Fund, Stockton University                      | July 2020      |

## SERVICE AND OUTREACH

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### Co-organizer

|                                                       |                     |
|-------------------------------------------------------|---------------------|
| Undergraduate Special Session, AWM Research Symposium | May 2025            |
| Experimental Mathematics Seminar, Rutgers University  | Fall 2024 - Present |
| EDGE Webinar Series, EDGE Foundation                  | Fall 2023 - Present |

**Co-founder** of the EDGE Webinar Series, EDGE Foundation 2023

**Member** of Department of Mathematics Climate Committee, Rutgers University 2023 - 2024

**Facilitator**, Cafe Con Leche, Research Talk hosted by Lathisms March 2022

**Co-founder and Co-president** of the Stockton University AWM Chapter 2020 - 2021

### Volunteer

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| Homework Completion Program, Atlantic City Library, NJ | March 2020 |
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| Networking/Speaker Series, Summer Math and Science Honors Academy | July 2019/2020 |
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**Proctor**, High School Math Mayhem Competition, Stockton University April 2019

**Service-Learning Coordinator** of the Alliance of EOF Students of NJ 2019 - 2020

**Referee** for Experimental Mathematics, PUMP Journal of Undergraduate Research

## MEMBERSHIPS

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|------------------------------------------------|-----------------------|
| Society for Industrial and Applied Mathematics | Jan. 2024 - Dec. 2025 |
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| American Mathematical Society | Oct. 2021 - Dec. 2025 |
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| Association for Women in Mathematics | July 2020 - July 2021 |
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| American Statistical Association | Nov. 2019 - Nov. 2020 |
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| Pi Mu Epsilon (Member, 2019-2020 Secretary) | Sept. 2019 - May 2021 |
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| Advancement of Chicanos/Hispanics and Native Americans in Science | July 2019 - July 2024 |
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## SKILLS

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### Technology

Proficient in  $\text{\LaTeX}$ , Sage, Maple and CoCalc. Some experience with HTML, Java and Python.

### Other

Fluent in English and Spanish, both written and verbal.